INTRODUCTION

Background

Internships play a crucial role in bridging academic learning and industry requirements. Educational institutions encourage students to participate in internships to gain practical exposure, develop technical skills, and improve employability. However, managing internship records for hundreds of students becomes difficult when handled through spreadsheets, emails, and paper documents.

Need for the System

Traditional methods suffer from data redundancy, lack of transparency, delayed communication, and difficulty in generating reports. A centralized Internship Tracking Portal addresses these issues by maintaining all records within a structured database system.

Scope of the Project

The system supports student registration, internship applications, progress monitoring, report submission, and faculty review. It provides a single platform for tracking internship activities.

Overview of DBMS Concepts Used

The project uses entities, attributes, relationships, primary keys, foreign keys, constraints, normalization, functional dependencies, joins, and query processing. These concepts ensure data consistency, integrity, and efficient retrieval.

Advantages

- Reduced manual effort
- Centralized record management
- Improved transparency
- Faster retrieval of information
- Better monitoring and analytics

Future Scope

Future enhancements may include AI-powered recommendations, placement integration.

LITERATURE SURVEY

Existing Systems

Many colleges maintain internship records using spreadsheets and email communication. Although easy to implement, these systems become difficult to manage as the number of students increases.

ERP-Based Systems

Certain educational ERP solutions provide internship tracking modules. These systems offer centralized access but are often expensive and contain unnecessary features.

Recruitment Portals

Online recruitment platforms support internship applications but are not designed for academic monitoring and report management.

Limitations of Existing Systems

• No centralized database
• Poor visibility into student progress
• Delayed communication
• Difficulty generating analytics
• Risk of data inconsistency

Comparative Analysis

Manual systems are low-cost but inefficient. ERP systems provide integration but increase complexity. The proposed Internship Tracking Portal balances simplicity, usability, and functionality.

Summary

The survey highlights the need for a dedicated academic internship management platform that supports monitoring, reporting, and transparency.

PROBLEM STATEMENT

Internships play a vital role in helping students gain practical knowledge and industry experience. However, in many educational institutions, internship-related activities are still managed through spreadsheets, emails, paper forms, and messaging applications. As the number of students increases, maintaining internship records using these methods becomes inefficient, time-consuming, and difficult to manage. Faculty members often face challenges in tracking student applications, monitoring internship progress, verifying submitted reports, and generating consolidated records.

The absence of a centralized system can lead to data redundancy, loss of important information, delayed communication, and difficulties in maintaining accurate records. Students may also find it challenging to keep track of application statuses, submission deadlines, and faculty feedback. Furthermore, generating department-level reports and analytics becomes a cumbersome task when data is scattered across multiple platforms.

To address these challenges, there is a need for a centralized and automated Internship Tracking Portal that can efficiently manage internship-related information. The proposed system provides a structured database-driven solution that enables students and faculty to access, update, and monitor internship records through a single platform. By automating key processes and ensuring data integrity, the system improves transparency, reduces manual effort, enhances communication, and provides a reliable mechanism for tracking internship activities throughout their lifecycle.

OBJECTIVES

To develop a centralized platform for managing and monitoring internship activities efficiently. The system aims to simplify the process of internship application tracking, report submission, and progress monitoring for both students and faculty. It seeks to reduce manual effort, improve transparency, and ensure accurate record management through a structured database system. The project also focuses on maintaining data consistency and integrity using DBMS concepts such as normalization, keys, and constraints. Additionally, it provides real-time activity tracking and analytics to support better decision-making and effective internship management within educational institutions.

REQUIREMENT ANALYSIS

HARDWARE REQUIREMENTS

• Processor : Intel Core i3 or higher

• RAM : 4 GB minimum (8 GB recommended)

• Storage : 500 MB free disk space

• Internet : Required for deployment and testing

SOFTWARE REQUIREMENTS

• Operating System : Windows 10/11, Linux, or macOS

• Frontend : React.js, Vite, Tailwind CSS

• Backend : Python 3.x, Flask

• Database : MySQL, MongoDB

• Version Control : Git, GitHub

• IDE : Visual Studio Code

• API Testing : Postman (Optional)

PYTHON DEPENDENCIES

• Flask

• Flask-CORS

• PyMySQL / mysql-connector-python

• python-dotenv

• PyJWT (if JWT authentication is implemented)

NODE.JS DEPENDENCIES

• Node.js (v18+ recommended)

• npm

• React

• React Router DOM

• Tailwind CSS

• Axios

DATABASE REQUIREMENTS

• **MySQL**

- Student Management
- Company Management
- Internship Records
- Applications Tracking

• **MongoDB**

- Activity Logs
- Analytics Data
- System Monitoring

BROWSER SUPPORT

• Google Chrome (Recommended)

• Microsoft Edge

• Mozilla Firefox

DEPLOYMENT REQUIREMENTS

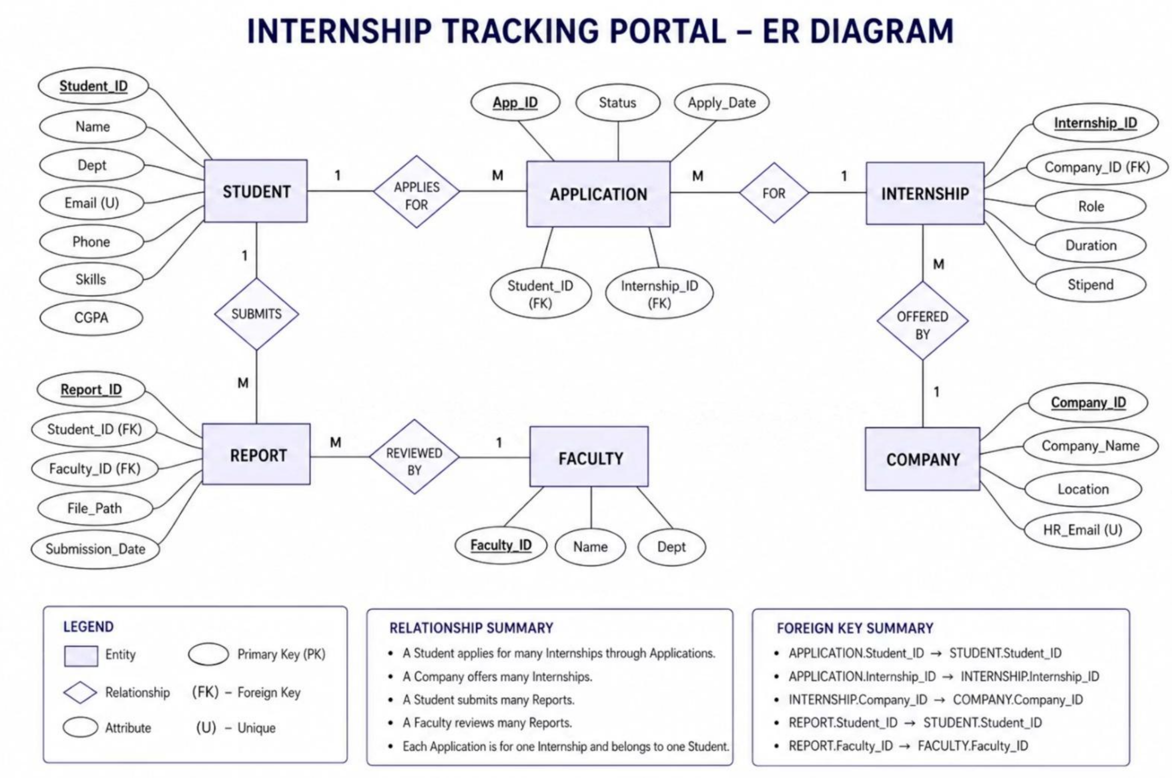
• GitHub Repository

• Render (Backend Hosting)

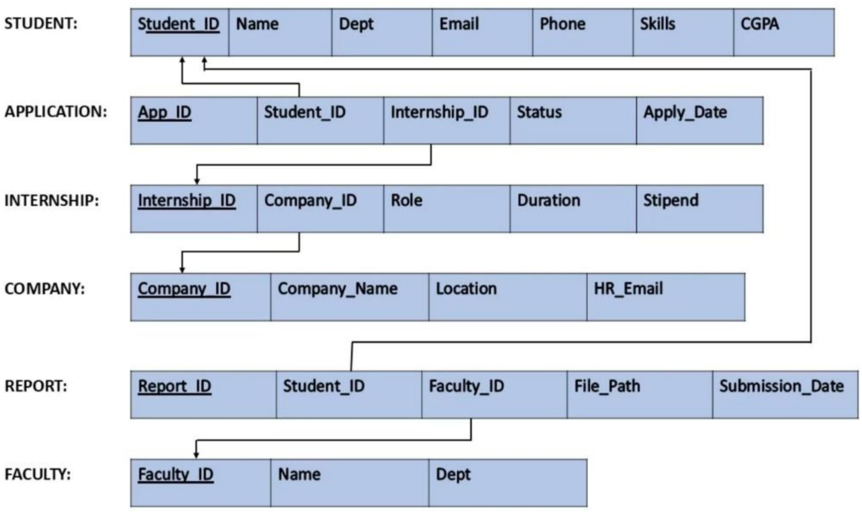
• Vercel / Netlify (Frontend Hosting)

DATABASE DESIGN

ER Diagram



Schema Diagram



IMPLEMENTATION

The Internship Tracking Portal is implemented as a full-stack web application using React.js, Flask, MySQL, and MongoDB. The frontend provides an interactive interface for students and faculty, while the backend handles business logic and database operations.

Frontend Implementation

The frontend is developed using React.js and includes:

- Student Registration and Login

- Browse Internships

- Apply for Internships

- Upload Reports

- Admin Dashboard

- Application Verification

- Report Review and Download

Backend Implementation

The backend is developed using Flask and provides REST APIs for:

- Authentication

- Internship Management

- Application Processing

- Report Upload/Download

- Analytics and Activity Tracking

Database Connectivity

MySQL

Used to store:

- Student Details
- Internship Information
- Applications
- Reports
- Faculty Data

MongoDB

Used for:

- Activity Logging
- Analytics Data
- Recent Activity Tracking

Modules Developed

Student Module

- Registration and Login
- Internship Search
- Internship Application
- Status Tracking
- Report Upload

Faculty/Admin Module

- Admin Login
- Application Verification

- Report Review

- Analytics Dashboard

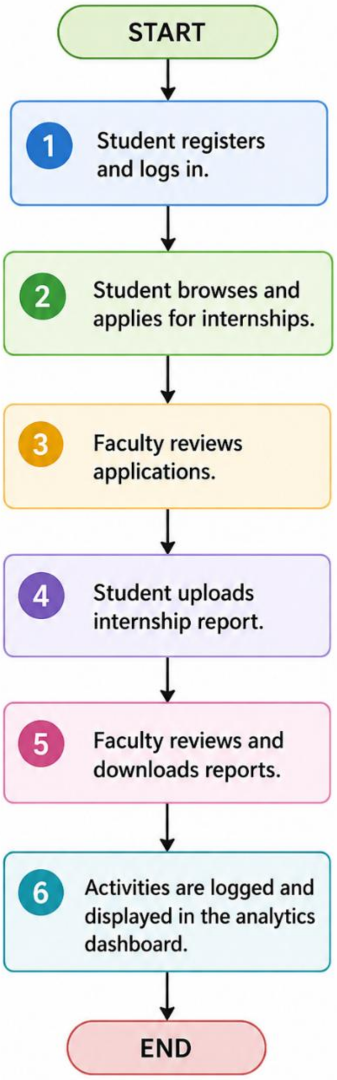
Report Management Module

- Report Upload

- Report Storage

- Report Download

System Workflow



RESULTS: SQL QUERIES AND OUTPUTS

CREATE TABLE queries:

```sql
CREATE TABLE STUDENT (
    Student_ID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    Dept VARCHAR(50) NOT NULL,
   Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15),
    Skills TEXT,
    CGPA DECIMAL(3,2),
    Password VARCHAR(255)
);
CREATE TABLE COMPANY (
    Company_ID INT PRIMARY KEY AUTO_INCREMENT,
    Company_Name VARCHAR(100) NOT NULL,
    Location VARCHAR(100),
    HR_Email VARCHAR(100) UNIQUE
);
```

```sql
CREATE TABLE STUDENT (
    Student_ID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    Dept VARCHAR(50) NOT NULL,
   Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15),
    Skills TEXT,
    CGPA DECIMAL(3,2),
    Password VARCHAR(255)
);
CREATE TABLE COMPANY (
    Company_ID INT PRIMARY KEY AUTO_INCREMENT,
    Company_Name VARCHAR(100) NOT NULL,
    Location VARCHAR(100),
    HR_Email VARCHAR(100) UNIQUE
);

CREATE TABLE FACULTY (
    Faculty_ID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    Dept VARCHAR(50),
    Email VARCHAR(100) UNIQUE,
```

UPDATE AND DELETE TABLE queries:

```sql
1. Update student's CGPA or details
UPDATE STUDENT
SET CGPA = 9.20, Skills = 'Python, React, Flask, Docker'
WHERE Student_ID = 1;

2. Update an application status
UPDATE APPLICATION
SET Status = 'Selected'
WHERE App_ID = 1;

3. Delete a specific report file entry
DELETE FROM REPORT
WHERE Report_ID = 6;

4. Delete an application entry
DELETE FROM APPLICATION
WHERE App_ID = 12;
```

JOIN queries:

```sql
CREATE OR REPLACE VIEW student_application_view AS
SELECT
    s.Student_ID,
    s.Name AS Student_Name,
    i.Role AS Internship_Role,
    c.Company_Name AS Company_Name,
    a.Status AS Application_Status,
    a.Apply_Date
FROM APPLICATION a
JOIN STUDENT s ON a.Student_ID = s.Student_ID
JOIN INTERNSHIP i ON a.Internship_ID = i.Internship_ID
JOIN COMPANY c ON i.Company_ID = c.Company_ID;
```

TRIGGERS:

```sql
CREATE TRIGGER prevent_invalid_cgpa
BEFORE INSERT ON STUDENT
FOR EACH ROW
BEGIN
    -- If CGPA provided and is outside allowed range, reject insert
    IF NEW.CGPA IS NOT NULL AND (NEW.CGPA < 0 OR NEW.CGPA > 10) THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid CGPA value';
    END IF;
END$$
DELIMITER ;
```

INSERT QUERIES

```sql
INSERT INTO INTERNSHIP (Company_ID, Role, Duration, Stipend)
VALUES
(1, 'Backend Developer Intern', '6 Months', 45000),
(1, 'Frontend Developer Intern', '3 Months', 30000),
(2, 'Cloud Engineer Intern', '6 Months', 50000),
(2, 'Data Analyst Intern', '4 Months', 35000),
(3, 'Software Engineer Intern', '6 Months', 40000),
(3, 'DevOps Intern', '5 Months', 42000),
(4, 'Full Stack Developer Intern', '6 Months', 38000),
(4, 'AI/ML Intern', '4 Months', 45000),
(5, 'Frontend Engineer Intern', '3 Months', 28000),
(5, 'Backend Engineer Intern', '6 Months', 36000),
(6, 'Java Developer Intern', '5 Months', 32000),
(6, 'UI/UX Developer Intern', '3 Months', 25000),
(7, 'Data Science Intern', '6 Months', 30000),
(8, 'Software Testing Intern', '4 Months', 22000);
```

SELECT queries:

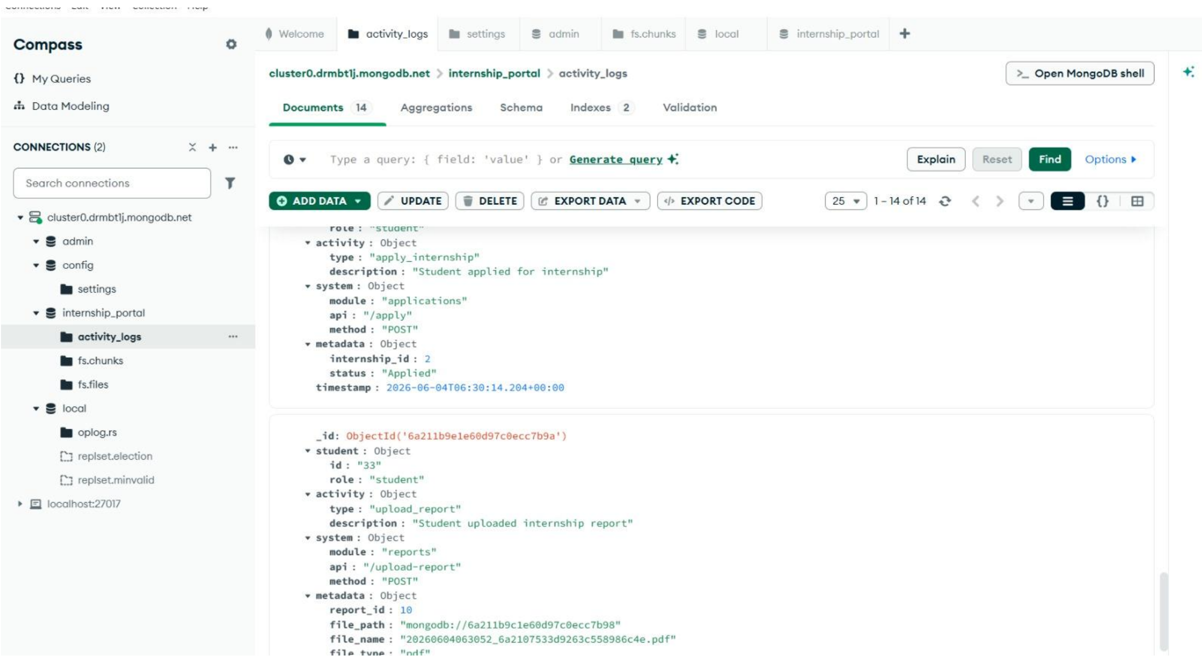
Limit to 1000 rows

```sql
SELECT * FROM railway.APPLICATION;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

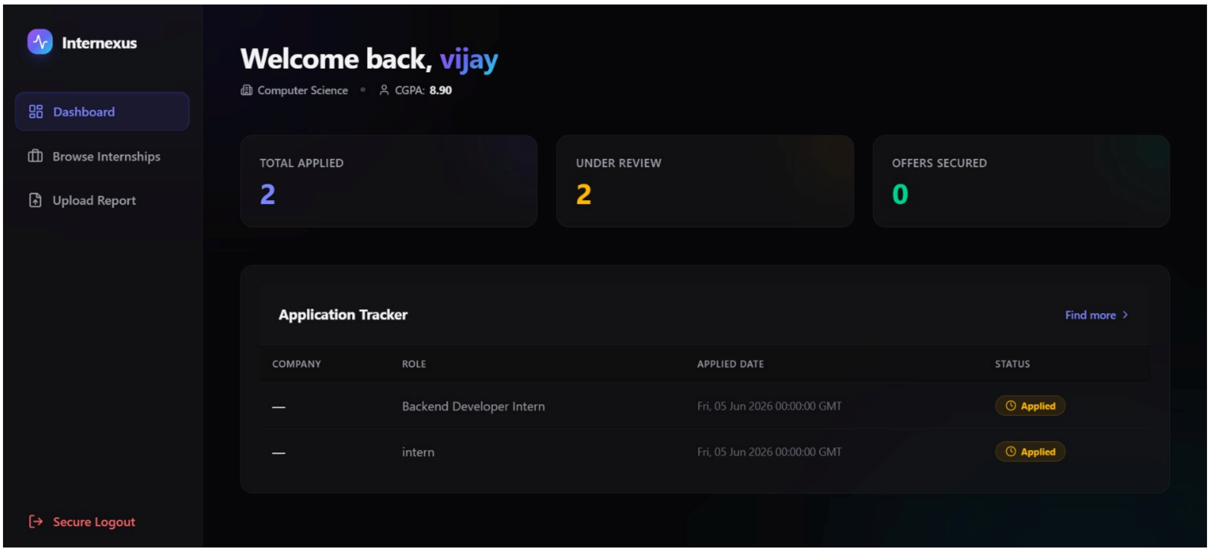
App_ID	Student_ID	Internship_ID	Status	Apply_Date
1	1	1	Applied	2026-05-01
2	1	7	Interview	2026-05-03
3	2	11	Selected	2026-05-02
4	2	4	Interview	2026-05-05
5	3	5	Rejected	2026-05-04
6	3	14	Applied	2026-05-07
7	4	1	Selected	2026-05-02
8	4	10	Interview	2026-05-08
9	5	8	Applied	2026-05-06
10	5	13	Interview	2026-05-09
11	6	8	Selected	2026-05-04
12	6	3	Interview	2026-05-10

NOSQL DATABASE:



DASHBOARD:

1.



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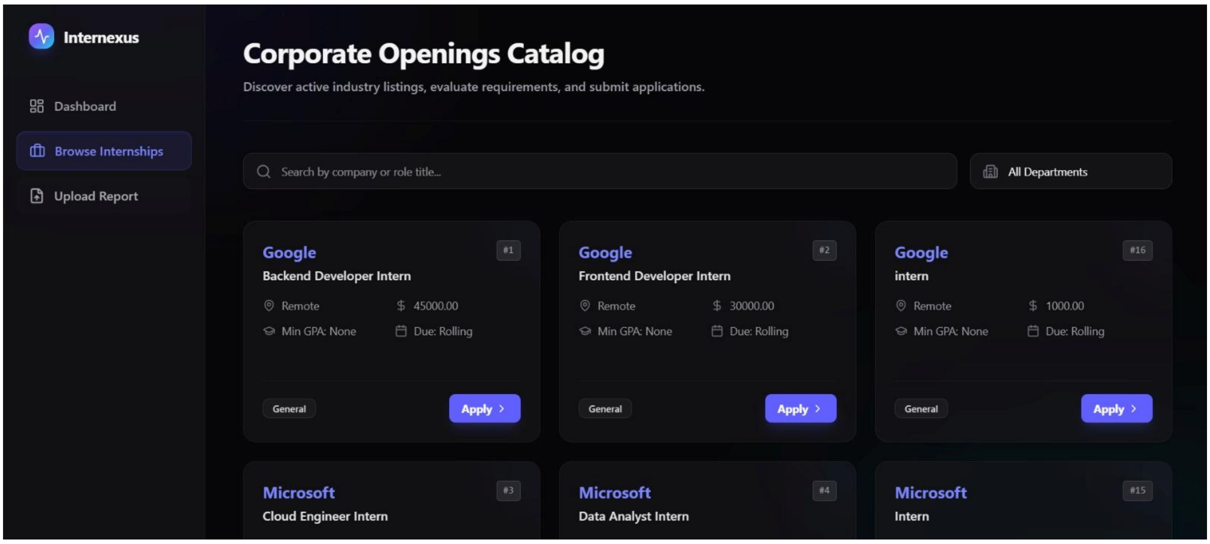
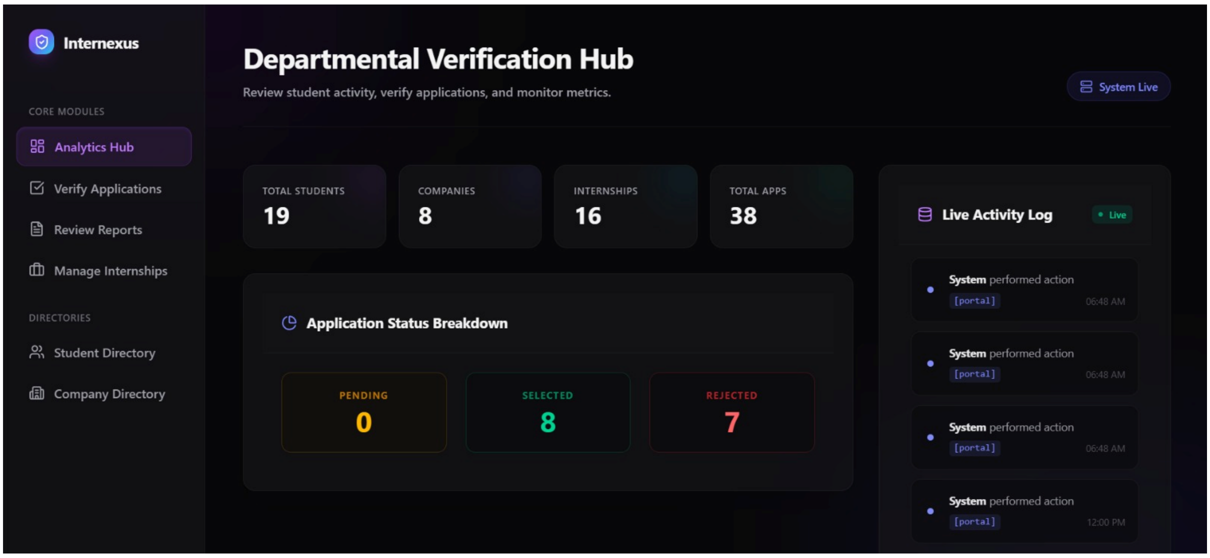
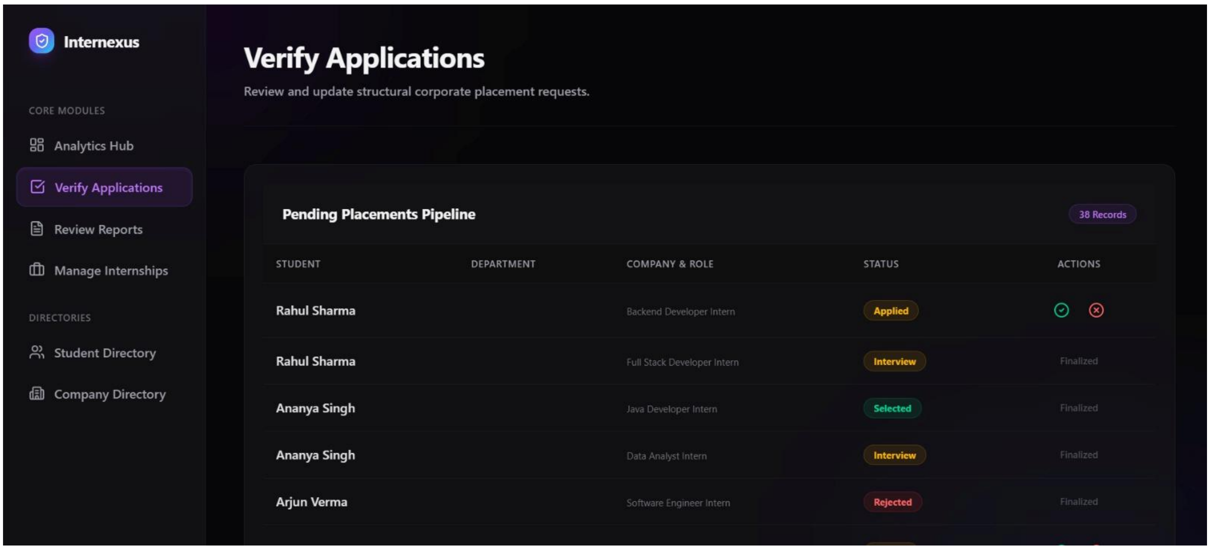

Internexus
Dashboard
Browse Internships
Upload Report
Corporate Openings Catalog
Discover active industry listings, evaluate requirements, and submit applications.
Search by company or role title...
All Departments
Google
Backend Developer Intern
#1
Remote
$ 45000.00
Min GPA: None
Due: Rolling
General
Apply
Google
Frontend Developer Intern
#2
Remote
$ 30000.00
Min GPA: None
Due: Rolling
General
Apply
Google
intern
#16
Remote
$ 1000.00
Min GPA: None
Due: Rolling
General
Apply
Microsoft
Cloud Engineer Intern
#3
Microsoft
Data Analyst Intern
#4
Microsoft
Intern
#15

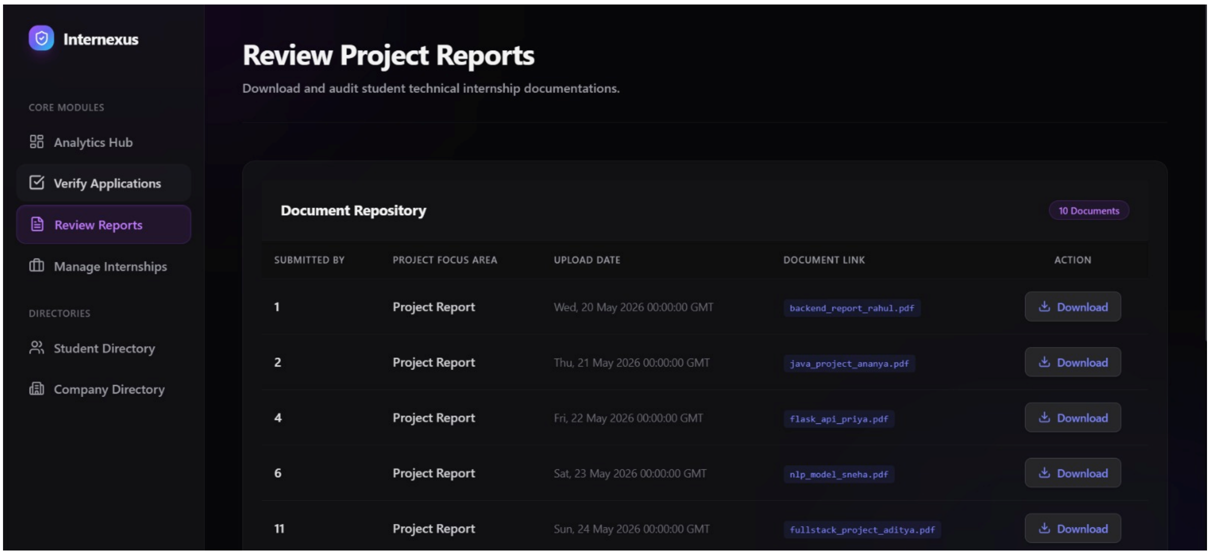
3.


Internexus
CORE MODULES
Analytics Hub
Verify Applications
Review Reports
Manage Internships
DIRECTORIES
Student Directory
Company Directory
Departmental Verification Hub
Review student activity, verify applications, and monitor metrics.
System Live
TOTAL STUDENTS
19
COMPANIES
8
INTERNSHIPS
16
TOTAL APPS
38
Application Status Breakdown
PENDING
0
SELECTED
8
REJECTED
7
Live Activity Log
Live
System performed action
[portal]
06:48 AM
System performed action
[portal]
06:48 AM
System performed action
[portal]
06:48 AM
System performed action
[portal]
12:00 PM

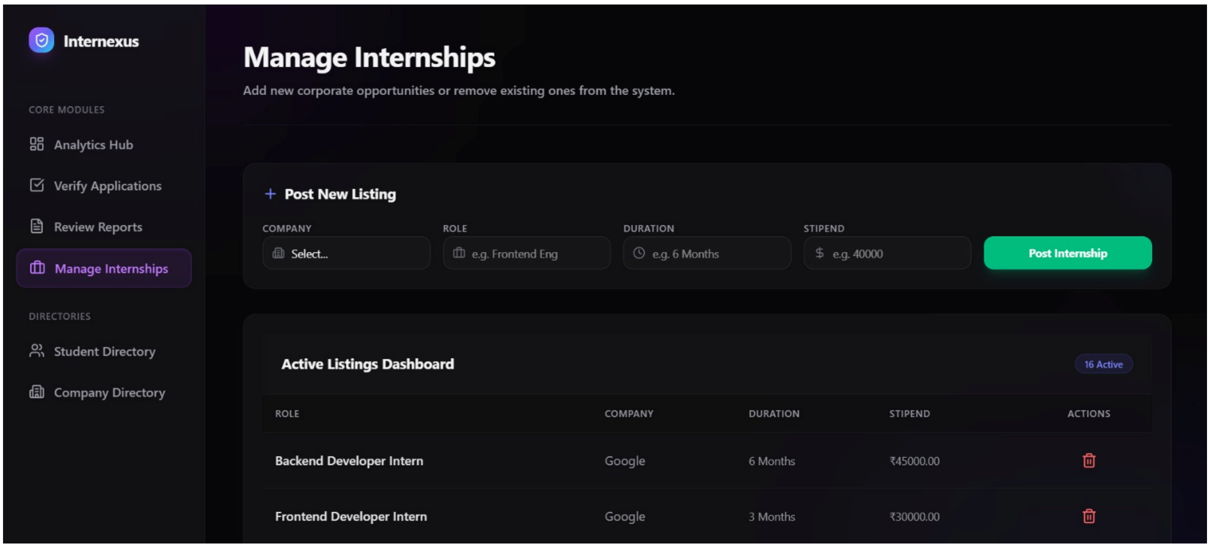
4.


Internexus
CORE MODULES
Analytics Hub
Verify Applications
Review Reports
Manage Internships
DIRECTORIES
Student Directory
Company Directory
Verify Applications
Review and update structural corporate placement requests.
Pending Placements Pipeline
38 Records
STUDENT
DEPARTMENT
COMPANY & ROLE
STATUS
ACTIONS
Rahul Sharma
Backend Developer Intern
Applied
Rahul Sharma
Full Stack Developer Intern
Interview
Finalized
Ananya Singh
Java Developer Intern
Selected
Finalized
Ananya Singh
Data Analyst Intern
Interview
Finalized
Arjun Verma
Software Engineer Intern
Rejected
Finalized

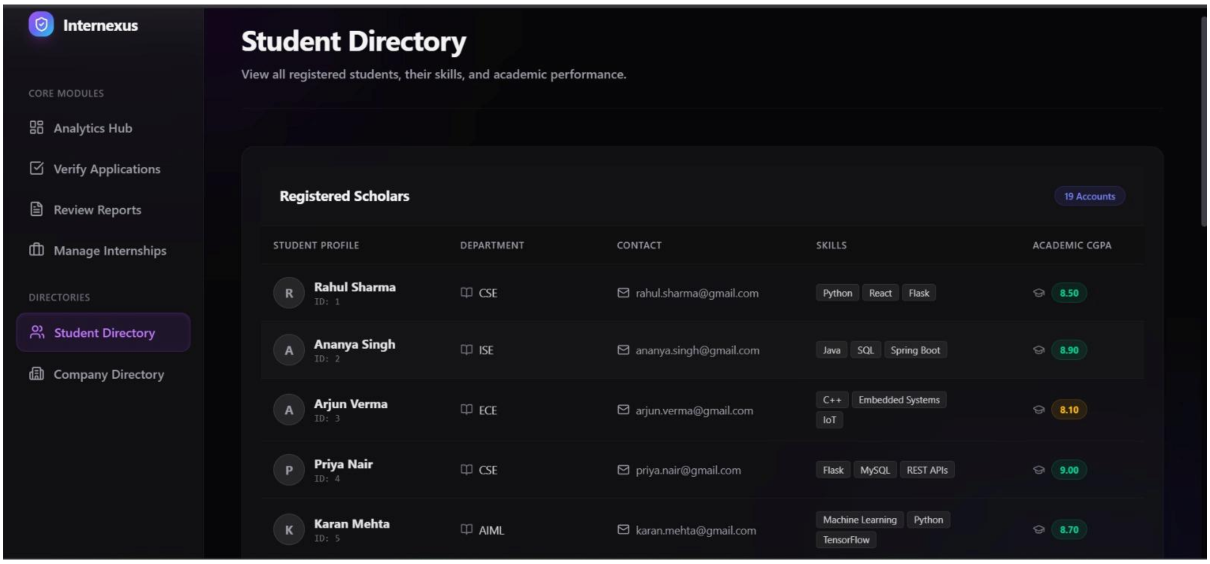
5.


Internexus
CORE MODULES
Analytics Hub
Verify Applications
Review Reports
Manage Internships
DIRECTORIES
Student Directory
Company Directory
Review Project Reports
Download and audit student technical internship documentations.
Document Repository
10 Documents
SUBMITTED BY
PROJECT FOCUS AREA
UPLOAD DATE
DOCUMENT LINK
ACTION
1
Project Report
Wed, 20 May 2026 00:00:00 GMT
backend_report_rahul.pdf
Download
2
Project Report
Thu, 21 May 2026 00:00:00 GMT
java_project_ananya.pdf
Download
4
Project Report
Fri, 22 May 2026 00:00:00 GMT
flask_api_priya.pdf
Download
6
Project Report
Sat, 23 May 2026 00:00:00 GMT
nlp_model_sneha.pdf
Download
11
Project Report
Sun, 24 May 2026 00:00:00 GMT
fullstack_project_aditya.pdf
Download

6.


Internexus
CORE MODULES
Analytics Hub
Verify Applications
Review Reports
Manage Internships
DIRECTORIES
Student Directory
Company Directory
Manage Internships
Add new corporate opportunities or remove existing ones from the system.
Post New Listing
COMPANY
Select...
ROLE
e.g. Frontend Eng
DURATION
e.g. 6 Months
STIPEND
e.g. 40000
Post Internship
Active Listings Dashboard
16 Active
ROLE
COMPANY
DURATION
STIPEND
ACTIONS
Backend Developer Intern
Google
6 Months
₹45000.00
Frontend Developer Intern
Google
3 Months
₹30000.00

7.


Internexus
CORE MODULES
Analytics Hub
Verify Applications
Review Reports
Manage Internships
DIRECTORIES
Student Directory
Company Directory
Student Directory
View all registered students, their skills, and academic performance.
Registered Scholars
19 Accounts
STUDENT PROFILE
DEPARTMENT
CONTACT
SKILLS
ACADEMIC CGPA
Rahul Sharma
ID: 1
CSE
rahul.sharma@gmail.com
Python
React
Flask
8.50
Ananya Singh
ID: 2
ISE
ananya.singh@gmail.com
Java
SQL
Spring Boot
8.90
Arjun Verma
ID: 3
ECE
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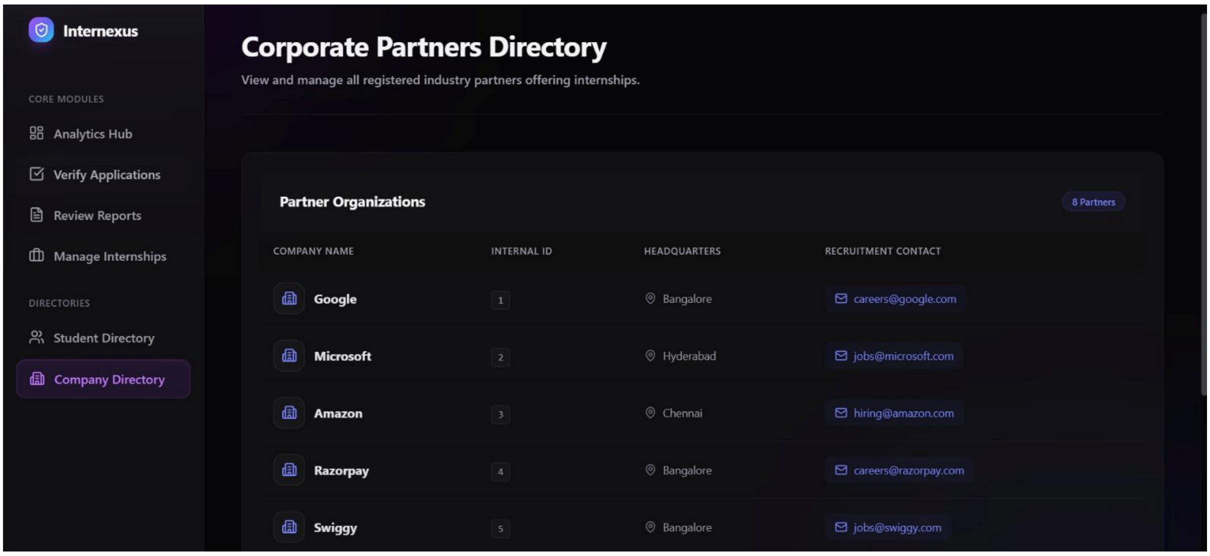
C++
Embedded Systems
IoT
8.10
Priya Nair
ID: 4
CSE
priya.nair@gmail.com
Flask
MySQL
REST APIs
9.00
Karan Mehta
ID: 5
AIML
karan.mehta@gmail.com
Machine Learning
Python
TensorFlow
8.70

8.


Internexus
CORE MODULES
Analytics Hub
Verify Applications
Review Reports
Manage Internships
DIRECTORIES
Student Directory
Company Directory
Corporate Partners Directory
View and manage all registered industry partners offering internships.
Partner Organizations
8 Partners
COMPANY NAME
INTERNAL ID
HEADQUARTERS
RECRUITMENT CONTACT
Google
1
Bangalore
careers@google.com
Microsoft
2
Hyderabad
jobs@microsoft.com
Amazon
3
Chennai
hiring@amazon.com
Razorpay
4
Bangalore
careers@razorpay.com
Swiggy
5
Bangalore
jobs@swiggy.com

CONCLUSION

The Internship Tracking Portal successfully addresses the challenges associated with manual internship management by providing a centralized, efficient, and user-friendly platform for students and faculty. The system streamlines the entire internship process, including application tracking, progress monitoring, report submission, and record management, thereby reducing the dependency on spreadsheets, emails, and paper-based documentation.

By utilizing a structured database design, the project ensures data consistency, integrity, and efficient retrieval of information. The implementation of DBMS concepts such as entity-relationship modeling, normalization, functional dependencies, minimal cover, attribute closure, primary keys, foreign keys, and integrity constraints has helped create a reliable and scalable database system. The integration of MySQL for relational data storage and MongoDB for the Live Log Tracking Stream further enhances the system's functionality and monitoring capabilities.

The project has also provided valuable practical experience in database design, web application development, and system integration. It demonstrates how database management principles can be applied to solve real-world problems in academic environments. Overall, the Internship Tracking Portal improves transparency, reduces administrative effort, enhances communication between stakeholders, and provides an effective solution for managing internship activities. The system can be further extended with advanced analytics, mobile support, and additional automation features in future versions.

REFERENCES

1. Abraham Silberschatz, Henry F. Korth, and S. Sudarshan, Database System Concepts, 7th Edition, McGraw-Hill Education, 2019.

2. MySQL Documentation, Available at: https://dev.mysql.com/doc/

3. Flask Documentation, Available at: https://flask.palletsprojects.com/

4. Internship Tracking Portal Project Repository, Available at: https://github.com/prashant25gourav/internship-tracking-portal

5.Deployed Project:

https://internship-frontend-1pcz.onrender.com